

| Question |     |      |     | Marking details                                                                                                                                                       | Marks Available |          |          |           |          |          |
|----------|-----|------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |      |     |                                                                                                                                                                       | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
| 3        | (a) |      |     | Helium nucleus                                                                                                                                                        | 1               |          |          | 1         |          |          |
|          | (b) |      |     | <u>Alpha</u> [particles] (1)<br>can't get through the <u>detector / air</u> (1)<br>Accept for 2 <sup>nd</sup> mark can't travel that far                              | 1               | 1        |          | 2         |          |          |
|          | (c) |      |     | [Gamma] would not be absorbed by smoke / only absorbed by lead (1)<br>Because they are very penetrating / they are low ionising / so the current would not change (1) | 1               | 1        |          | 2         |          |          |
|          | (d) | (i)  | I   | 95 (1)<br>0 (1)                                                                                                                                                       | 1               | 1        |          | 2         |          |          |
|          |     |      | II  | 94 [protons]                                                                                                                                                          |                 | 1        |          | 1         |          |          |
|          |     |      | III | 147 [neutrons]                                                                                                                                                        |                 | 1        |          | 1         |          |          |
|          |     | (ii) |     | Beta particle                                                                                                                                                         | 1               |          |          | 1         |          |          |
|          |     |      |     | <b>Question 3 total</b>                                                                                                                                               | <b>5</b>        | <b>5</b> | <b>0</b> | <b>10</b> | <b>0</b> | <b>0</b> |